

# Jongmin Lee

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| POSITION            | <b>Ph.D Candidate</b><br><b>Department of Computer Science and Engineering</b><br><b>Pohang University of Science and Technology (POSTECH)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                          |  |
| CONTACT INFORMATION | Computer Vision Laboratory, E2 302, Dept. of CSE, POSTECH<br>77 Cheongam Rd, Nam-gu,<br>Pohang, Gyeongbuk, 37673,<br>Republic of Korea<br>Phone: (+82) 54 279 2931<br>Mobile: (+82) 10 8242 3070<br>e-mail: <a href="mailto:ljm1121@postech.ac.kr">ljm1121@postech.ac.kr</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Homepage: <a href="#">blog</a> , <a href="#">google scholar</a> , <a href="#">github</a> |  |
| CITIZENSHIP         | The Republic of Korea                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                          |  |
| BIRTHDAY            | November 21th, 1994.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                          |  |
| RESEARCH INTERESTS  | Computer vision and deep learning.<br>Specific research interests include: <b>image matching</b> , <b>deep local feature extraction</b> , semantic correspondence, self-supervised learning, equivariant representation learning, geometric deep learning, extremely low-light image enhancement, and burst image enhancement/restoration.                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                          |  |
| EDUCATION           | <b>Pohang University of Science and Technology (POSTECH)</b> , Pohang, Korea<br><i>Ph.D Student, Dep. of Computer Science and Engineering (CSE)</i> <b>Sep 2018 – now</b> <ul style="list-style-type: none"><li>• Advisor: <a href="#">Prof. Minsu Cho</a></li></ul><br><b>Graz University of Technology (TU Graz)</b> , Graz, Austria<br><i>Visiting Student</i> <b>Sep 2016 – Feb 2017</b><br><br><b>Pohang University of Science and Technology (POSTECH)</b> , Pohang, Korea<br><i>B.S., Dep. of Industrial and Management Engineering (IME)</i><br><i>B.S., Dep. of Computer Science and Engineering (CSE)</i> <b>Mar 2013 – Aug 2018</b> <ul style="list-style-type: none"><li>• Double Major</li></ul>                                                                         |                                                                                          |  |
| INDUSTRY EXPERIENCE | <b>Meta Reality Labs.</b> , Burlingame, California, U.S.A.<br><i>Ph.D Research Scientist Intern</i> <b>Jul 2023 – Oct 2023</b> <ul style="list-style-type: none"><li>• Learning-based analytics research on human behavior. (Upcoming)</li></ul><br><b>Fast campus.</b> , Seoul, South Korea<br><i>Computer Vision Lecturer</i> <b>Mar 2022 – Aug 2022</b> <ul style="list-style-type: none"><li>• Lecturing on computer vision from classical to state-of-the-art deep learning approaches.</li></ul><br><b>Vuno Inc.</b> , Seoul, South Korea<br><i>Front-end Developer</i> <b>Jun 2017 – Aug 2017</b> <ul style="list-style-type: none"><li>• Developed the client/front-end interface for an AI-based software for diagnosis of major abnormalities from a chest X-ray.</li></ul> |                                                                                          |  |

INTERNATIONAL  
PUBLICATIONS

**Jongmin Lee**, Min Jung Lee, Sanghyun Kim, Woohyeok Kim, Jaesung Rim, Sunghyun Cho, Minsu Cho, “Burst Image Enhancement with Non-Uniform Exposures,” *Under review*.

**Jongmin Lee**, Sanghyun Kim, Min Jung Lee, Minsu Cho, “Multiply Degraded Burst Image Enhancement via Robust Base Frame Selection,” *Under review*.

**Jongmin Lee**, Byungjin Kim, Seungwook Kim, Minsu Cho, “Learning Rotation-Equivariant Features for Visual Correspondence,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

**Jongmin Lee**, Byungjin Kim, Minsu Cho, “Self-Supervised Equivariant Learning for Oriented Keypoint Detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

**Jongmin Lee**, Yoonwoo Jeong, Minsu Cho, “Self-supervised Learning of Image Scale and Orientation Estimation,” in *Proceedings of the 32nd British Machine Vision Conference (BMVC)*, 2021.

**Jongmin Lee**, Yoonwoo Jeong, Seungwook Kim, Juhong Min, Minsu Cho, “Learning to Distill Convolutional Features Into Compact Local Descriptors,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2021.

Juhong Min, **Jongmin Lee**, Jean Ponce, Minsu Cho, “Learning to Compose Hypercolumns for Semantic Visual Correspondence,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2020.

Juhong Min, **Jongmin Lee**, Jean Ponce, Minsu Cho, “SPair-71k: A Large-scale Benchmark for Semantic Correspondence,” arXiv preprint, 2019.

Juhong Min, **Jongmin Lee**, Jean Ponce, Minsu Cho, “Hyperpixel Flow: Semantic Correspondence with Multi-layer Neural Features,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.

Paul Hongsuck Seo, **Jongmin Lee**, Deunsol Jung, Bohyung Han, Minsu Cho, “Attentive Semantic Alignment with Offset-Aware Correlation Kernels,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018.

RESEARCH  
PROJECTS

**Kakao Brain** Nov. 2021 - Jul. 2023  
Efficient equivariant representation learning in deep neural networks.

**Samsung Advanced Institute of Technology (SAIT)** Nov. 2022 - Jul. 2023  
Non-uniformly exposed burst image restoration using robust base frame selector.

**Samsung Advanced Institute of Technology (SAIT)** Nov. 2021 - Oct. 2022  
Burst image enhancement in an extremely degraded environment by noise, blur and shift.

**Samsung Advanced Institute of Technology (SAIT)** Nov. 2020 - Oct. 2021  
Motion-aware burst image enhancement under extremely low-light conditions.

PROFESSIONAL  
ACTIVITIES

**Reviewer of international conferences**  
Neural Information Processing Systems (NeurIPS) 2023  
International Conference on Computer Vision (ICCV) 2023

International Conference on 3D Vision (3DV) 2022  
 European Conference on Computer Vision (ECCV) 2022  
 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2022, 2023  
 British Machine Vision Conference (BMVC) 2021  
 Winter Conference on Applications of Computer Vision (WACV) 2021, 2022, 2023  
 International Conference on Machine Vision Applications (MVA) 2021, 2023  
 International Conference on Pattern Recognition (ICPR) 2020

#### Reviewer of international journals

IEEE Transactions on Pattern Analysis and Machine Intelligence (2023)  
 International Journal of Computer Vision (2023)  
 IEEE Transactions on Image Processing (2022, 2023)  
 Pattern Recognition (2022)  
 The Visual Computer (2022)

#### Teaching Assistant

|                                                            |                      |
|------------------------------------------------------------|----------------------|
| Artificial Intelligence & Data Science (CSED537) @ POSTECH | Spring semester 2023 |
| Introduction to Deep Learning (CSED490W) @ POSTECH         | Fall semester 2019   |
| POSCO AI expert training course                            | July 2019 – Aug 2019 |
| Automata & Formal Languages (CSED341) @ POSTECH            | Fall semester 2018   |

#### HONOURS AND AWARDS

BK21 outstanding paper award, POSTECH CSE, 2022.

Global Ph.D fellowship, *National Research Foundation of Korea (NRF)*, 2019 – Now.

Excellent research award, *Undergraduate Research Program, POSTECH Computer Science Engineering Dep.*, 2018.

SK Hynix scholarship, *SK Hynix Fellowship Program, POSTECH*, 2015.

#### INVITED TALKS

“Hyperpixel Flow: Semantic Correspondence with Multi-layer Neural Features“, *ICCV 2019 Paper Day with Naver*, Hotel Andaz Gangnam, Seoul, Korea, October 2019.

“Where is semantic correspondence? - The general image matching problem in deep learning era“, *Hyundai Motors AIR Lab(Artificial Intelligence Research Lab) seminar*, Pohang, Korea, September 2019.

“Semantic Alignment - Find Semantic Dense Correspondence,” *Naver corp.*, Pangyo, Korea, October 2018. [Youtube link \(Korean\)](#)

LANGUAGE SKILLS Korean(native), English(fluent)

REFEREES *Available on request.*